labor income co-moves with equities. Whether you should hold more or fewer equities as you approach retirement crucially depends on whether you are a stock or a bond.

4. Retirement

4.1. REPLACEMENT RATES

In reforming Rhode Island's pension plan, Raimondo started at the end and asked what would be optimal for an individual to do at retirement. Then, with a secure retirement in mind, she asked how a retirement plan should be designed to get there. Raimondo is not an economist (she graduated from Yale Law School), but she employed the same solution that an economist would have used. Economists solve life-cycle models by starting at the end of the problem, finding its solution, and then stepping back one period to solve the younger person's problem given the optimal solution of the older retiree. This procedure is called *dynamic programming* (see chapter 4).

In examining Rhode Island's pension plan, Raimondo's team found some cases where some individuals had replacement rates well above one. However, while most working Americans can rely on Social Security to establish a (minimal) base for their replacement income, over 50% of teachers and many public safety employees in Rhode Island's plan did not participate in Social Security.

Economists believe that the replacement rate, the amount of income in retirement as a fraction of pre-retirement income, should be less than 100% for most people. There are a number of justifications for this advice.[37] At retirement, households usually face lower tax rates. Social Security benefits, for example, are more lightly taxed than regular wages, or not taxed at all, in some states. Retirees sometimes downsize their housing (but many do not, see below), leading to lower property taxes. But most important, households do not need to save for retirement—they are now in retirement!

The World Bank, the Employee Benefit Research Institute, and other academic and policy institutions advocate a target replacement rate of 75% to 80%.[38] Scholz and Seshadri (2009) compute optimal replacement rates using a life-cycle model and find median optimal target rates of 75% for married couples and 55% for singles. There is, however, a large amount of dispersion as many other factors, like the number of dependent children, the level of income, and other household-specific considerations enter the computation.

Some people prefer other patterns of retirement income. Some prefer their income never to fall. In fact, for many institutions with their own life cycles, this is

[37] See Scholz and Seshadri (2009).

[38] The OECD estimates the average replacement rate is 60% across all OECD countries. See OECD, Pensions at a Glance 2011: Retirement-Income Systems in OECD and G20 Countries.